Statistics I - Lecture notes (sample for automated tests)

Chapter 1: Hypothesis tests
A hypothesis test compares a null hypothesis H0 with an alternative hypothesis H1.
The p-value is the probability of data at least as extreme as observed, assuming H0 is true.
Type I error: rejecting H0 although it is true. Type II error: keeping H0 although it is false.

Chapter 2: Linear regression
Simple linear regression fits y = a + b x by least squares.
The coefficient of determination R squared measures the share of explained variance.